MC 501 Inferencing Data in R rubric

100 points. Due Monday, December 7, 11:59 PM.

How this is scored. Add the points from each row; the total is out of 100. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Attempts are unlimited, so you can revise an early submission right up to the deadline.

- Go deep on one test instead of wide across several. Further tests earn nothing extra, and three tests reported thinly score below one reported with its assumptions, diagnostics, effect size and power.
- Choosing from your variable types: two categorical variables take a chi-square test of independence; one two-group categorical variable against one continuous variable takes an independent t-test; two continuous variables take a Pearson correlation.

Inferencing Data in R, 100 points

Criterion	Exemplary	Proficient	Developing	Beginning
Hypothesis 8 points	**8 points.** Stated in plain English before any code, naming the two variables and the relationship you expect.	**6 to 7 points.** Stated before the code, but one variable is vague.	**4 to 5 points.** Stated after the results, or as a topic instead of a relationship.	**0 to 3 points.** Missing.
Test selection and the rejected alternative 15 points	**15 points.** A test that fits your variable types, justified against one alternative you rejected and why.	**11 to 14 points.** The right test, but the rejected alternative or the reason is missing.	**8 to 10 points.** A test that does not fit the variable types.	**0 to 7 points.** No test chosen.
Assumptions and diagnostics 20 points	**20 points.** Each assumption of the test checked, with the diagnostic shown, and what you did about any that fail.	**15 to 19 points.** Checked and shown, but nothing said about one that fails.	**10 to 14 points.** Assumptions listed or asserted ("the data are normal") without being shown.	**0 to 9 points.** Missing.
APA reporting 15 points	**15 points.** The test runs in your document and is reported in APA format with the statistic, degrees of freedom, p and n.	**11 to 14 points.** One of the four is missing.	**8 to 10 points.** The statistic and p only, or numbers pasted in instead of produced by the code.	**0 to 7 points.** Not reported.
Effect size 15 points	**15 points.** Expressed in the units of your data as well as by its conventional label.	**11 to 14 points.** The label only.	**8 to 10 points.** Computed but not explained, or the wrong measure for the test.	**0 to 7 points.** Missing.
Power 15 points	**15 points.** Reports what effect this sample could detect, and what that means for a null result.	**11 to 14 points.** The detectable effect, without what it means for a null result.	**8 to 10 points.** A power figure with no link to your own design (a generic eighty percent).	**0 to 7 points.** Missing.

Criterion	Exemplary	Proficient	Developing	Beginning
Significance against size, and the limit 12 points	**12 points.** A paragraph that would survive a reviewer, separating statistical significance from practical size, and one paragraph on what the result does not let you claim.	**9 to 11 points.** Separates the two, but the paragraph on what the result does not license is generic.	**6 to 8 points.** Treats a small p as a large effect, or the paragraph on what it does not license is missing.	**0 to 5 points.** Missing.

Before you submit

Inferential Check recomputes a chi-square or a t test from your own table or group figures and returns the statistic, the degrees of freedom, p, the effect size, the assumptions your data may be breaking, and the APA sentence to compare against what you wrote. It is a second opinion on your R output, not a replacement for it. It does not grade, and nothing you enter leaves your machine.